

Realistic Interactive Consumer Simulation Grounded in Household Evidence for Food-Waste Behavior-Change Assistance

How well do scenarios generated by large language models (LLMs) preserve household evidence?

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MAIN FINDINGS

Sampling target categories improves population fit.

Sequential elaboration reduces measured target retention.

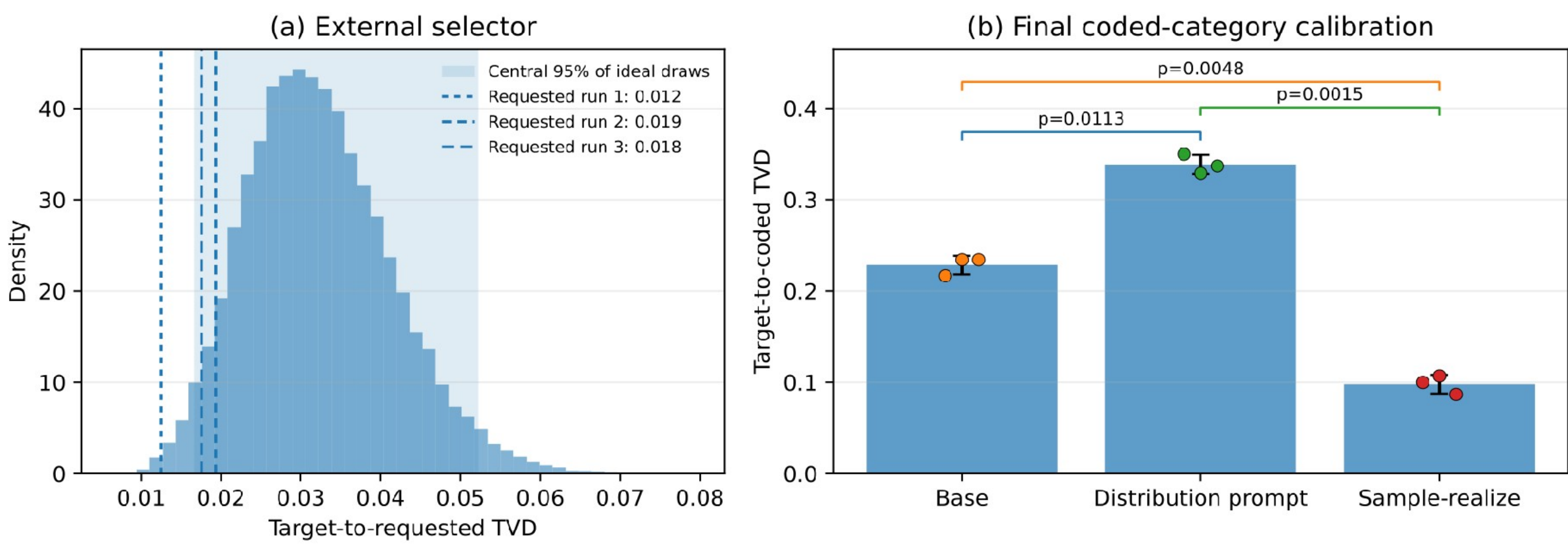
Population-level agreement and fidelity within individual scenarios need separate checks.

Food-waste assistance needs concrete accounts of household decisions, from planning to leftovers. LLMs can connect fragmented survey and diary evidence into scenarios. We test whether those scenarios preserve the supplied evidence.

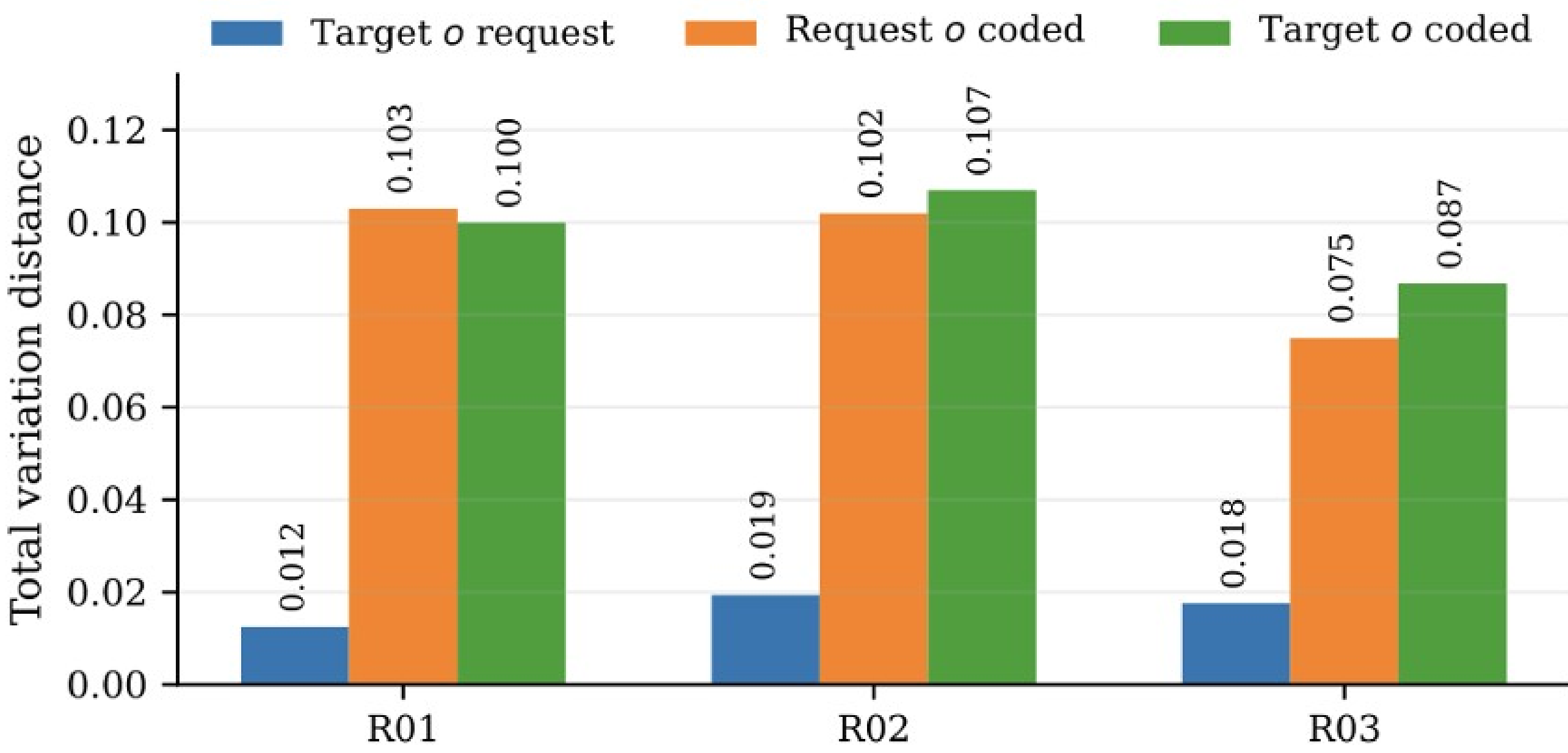
01 Matching population distributions

We generated 1,000 scenarios per corpus using three methods: unconstrained generation, the full target distribution in the prompt, or a sampled target category. Each method ran three times. The same classifier coded food categories in every condition.

Mean total variation distance (TVD; lower is closer):
0.098 sampled, 0.228 unconstrained, 0.339 distribution prompt.



Pre-sampling gave the closest final distribution. Dots show individual generation runs.



Selected categories were close to target. Further mismatch appeared after generation and coding.

Calibration does not ensure broader realism

Better category calibration did not consistently improve food identity, linguistic or narrative diversity, household-process variation, or human-perceived realism.

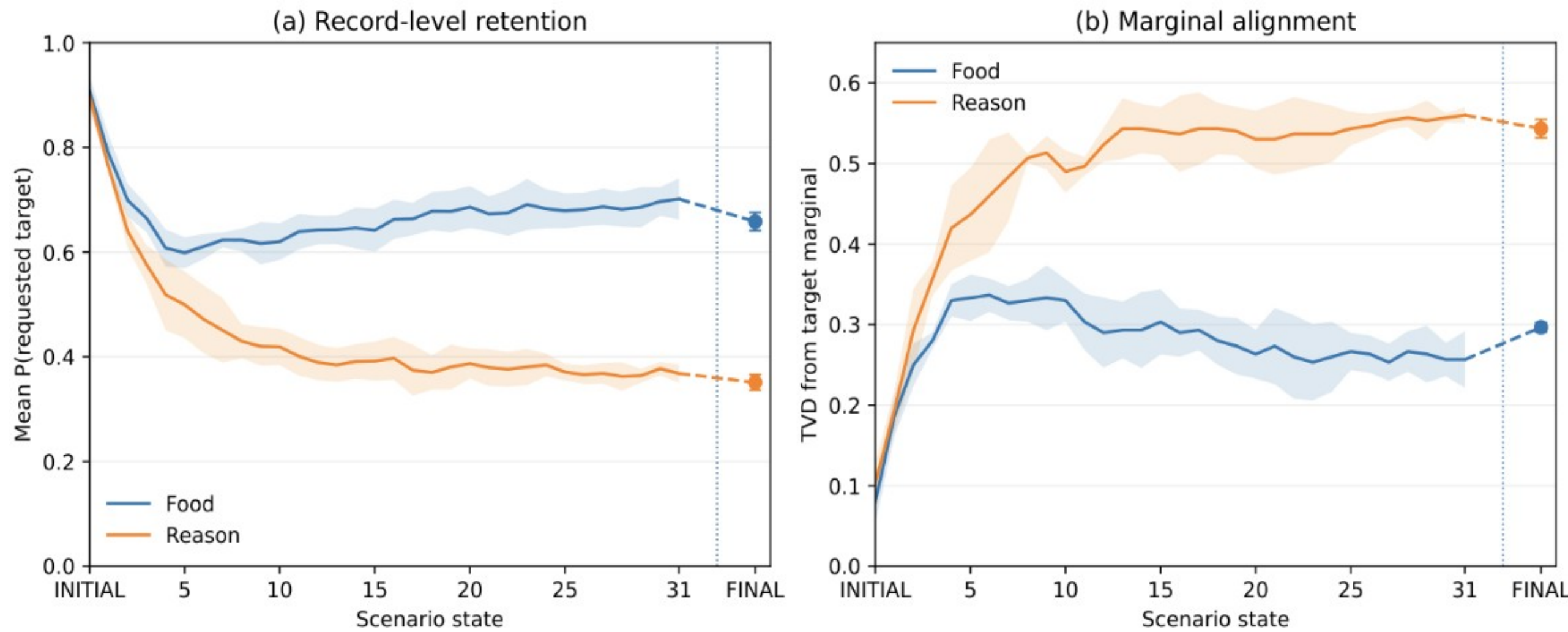
02 Following assigned information

We assigned a food category and disposal reason to each of 100 cases, using Finnish household diary shares. The same 31 questions were answered in one call or sequentially, with each answer added to the scenario. A classifier coded each stage without seeing the assigned targets.

Final match to assigned targets and distributional distance

Condition	Food match	Reason match	Food TVD	Reason TVD
One-shot	0.820	0.450	0.170	0.360
Sequential	0.657 ± 0.015	0.353 ± 0.015	0.297 ± 0.006	0.543 ± 0.012

Match: higher is closer. TVD: lower is closer.



Most change occurred during elaboration, with stronger change in the assigned reason than in the food category.

Sequential construction had lower target match and greater distributional distance than one-shot generation. These are classifier-based measures of mismatch.

Example: the primary reason label changes

Assigned targets: home-cooked food; over-prepared

Stage	Generated narrative excerpt	Coded reason
Initial	"...far exceeding the number of servings needed for tonight's dinner."	Over-prepared
Turn 12	"...leftover lasagna had gone sour and was discarded."	Spoiled or mouldy

Over-preparation and later spoilage can coexist. A change of primary label alone does not establish information loss.

03 Planned study: evaluating richer household scenarios

Multiple compatible labels

Allow several food and reason labels at population and case levels. Human checks can distinguish contradiction, omission and compatible elaboration.

Household profile information

Show, hide or alter age, household size and meal planning. Compare profile constructions with the same published margins but different attribute combinations.

Survey answers held out as targets

Ask about "best before" behaviour and "use by" beliefs without giving target answer distributions. Compare the generated answers with the Danish survey.

Sources

Study 1: "Calibrating Household Food-Waste Scenario Generation: From Category Marginals to Narrative Complexity."
Study 2: "Diagnosing Empirical Target Retention in Multi-Turn LLM Scenario Construction for Household Food Waste."
Finnish diary targets: Silvennoinen et al. (2014), British Food Journal 116(6):1058–1068.
Planned study: Norstat for the Danish Agriculture & Food Council, October 2024 (n = 1,512).